UNIVERSITÀ COMMERCIALE LUIGI BOCCONI

Bachelor of Science in International Economics and Management

Data Analytics within the National Basketball Association: an Outlook of Historical Trends and a Model-Based Assessment of their Implications on Roster Construction

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Abstract

During the last two decades, the sector of data analytics has seen a significant surge in applications within sports, primarily as a means to achieve superior performance and gain an advantage over competitors. Moreover, in recent years sport organizations have started to invest significant amounts of capital towards the development of their own data analytics departments. This constantly-increasing focus on data has its roots on the importance of the insights that can be derived from its accurate analysis. As a consequence, more and more sport organizations rely on a heavy use of data – ranging from simple shot-charts to more advanced player-movement tracking systems – to develop insights that can contribute to shaping their offensive and defensive strategies. In turn, this allows to achieve superior results if compared to those of their opponents: deriving valuable insights from in-game statistics is crucial to obtain a competitive advantage and be better-suited to succeed. Subsequently, the following final work will apply the use of data analytics to the context of sports by focusing on the National Basketball Association (NBA). As one of the largest sport leagues in the world in terms of revenue – it ranks second with a total revenue of $11 billion – the NBA and the teams that compose it invest significantly in the use of data analytics, not only in terms of human-capital but also in terms of development and implementation of sophisticated infrastructure. This allows for vast amounts of data that are available directly from the Association's website. By leveraging programming language R and its data analytics and visualization capabilities, this final work pursues an analysis of historical players' and teams' statistics in order to assess whether it is possible to identify changes in the way basketball is played. Subsequently, efforts are made to develop a clustering model that is able to group players according to similarities among their in-game statistics, without regards for the player's classical position. This analysis allows to identify subsets of players whose characteristics and skill-sets are similar, thus contributing to the development of five new, alternative positions in which to partition current NBA players. To conclude, this final work presents one of the several applications of data analytics processes by focusing on NBA basketball. This is achieved primarily through visualizations of historical trends and modeling of players' statistics. The analysis contributes to casting light upon the importance of data within sports organizations. Data, altogether with the insights that can be derived from its analysis, represents a significant source of competitive advantage over opponents, thus leading to superior player and team performance.

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1 Introduction

1.1 The Advent of Analytics in the NBA

In the last two decades, and even more so during recent years, the sector of data analytics has experienced and is continuing to experience a significant surge in applications within sport organizations and teams. Indeed, several organizations competing in the major sport leagues worldwide (*e.g*: NBA, NFL, Serie A, La Liga, etc.) rely heavily on the use of data to guide management-level decisions through the exploitation of valuable insights derived from the data collected directly from the playing field.

Indeed, thanks to the development of sophisticated tracking technologies, altogether with powerful statistical modeling capabilities, sport organizations are constantly improving their data collection and analysis processes, especially as leagues are embracing data-driven decision-making. Subsequently, organizations can derive valuable insights to improve their decision-making processes in a vast series of sectors, such as injury prevention, in-game strategies, roster composition and even player compensation.

Nonetheless, despite the significant relevance of data, especially in terms of improved decision-making processes, the role played by analytics within sports is not appreciated by all the parties taken into consideration. Indeed, it is fairly common for players — especially former players that played before the advent of analytics [1] — to contest the reliance of organizations on such a massive use of data. Some argue that statistics cannot fully grasp all the processes that take place on a playing field, and thus that the insights derived from data do not always represent the optimal solutions and responses to in-game scenarios. On the other hand, current players are starting to increasingly embrace the adoption of analytics and statistics, particularly as a method to model their playing styles to achieve the highest possible efficiency. This often also results in longer playing careers.

The use of analytics in sports, especially within basketball and the National Basketball Association (NBA), is now as high as ever, with each organization setting up and structuring their own analytics division with the hope of achieving a competitive advantage over the other teams.

1.2 The Impact of Analytics

The direct consequence of such reliance on data and the derived insights can be identified as a significant shift in tendencies and playing styles – especially if compared to the late 1990s. Indeed, focusing on insights derived from the collected data allowed teams to identify the most appropriate and satisfactory decisions to in-game events. This in turn implies a major focus on executing actions with the highest possible expected value – even if means making decisions which do not seem the best option.

The corresponding result is that the game of basketball, as played in the NBA, was subject to major changes in playing style. First and foremost, the game is now played at a higher speed, since a greater number of offensive possessions allows for more scoring opportunities. Additionally, especially in the last fifteen years, there has been a radical increase in reliance on the three-point shot. Additionally, another major consequence of such changes is directly reflected on players and rosters. Indeed, especially due to the advent of the three-point shot, there has been a major restructuring process that redefined the five playing positions within a team. Indeed, if in the past the distinction between one playing position and another was clearly defined, now that is no longer the case: the change in the way basketball is played caused players to adapt their playing style and competencies to match the new set of skills required to be successful in the modern NBA. As a result, the classical distinction among the five positions is no longer crystal clear; this contributed to the emergence of new, hybrid positions whose competencies are not fixed, but are rather built on the player's strengths and that can often overlap between several of the five classical positions.

Ultimately, since the late 1990s, the game of basketball in general, and specifically the National Basketball Association, have experienced a significant surge in appreciation worldwide. This in turn translated to an increase in the demand for basketball media consumption, both nationally and worldwide. Subsequently, the NBA has seen a rapid and significant increase in its reach – and revenues – that contributed to strengthen its bargaining power in media deals negotiations. This matter is particularly relevant now: the National Basketball Association recently concluded (June, 2024) a massive, 11-year, $76 billion media agreement in partnership with The Walt Disney Company, Amazon and NBCUniversal [12]. While being beneficial to the NBA as an organization, media deals also benefit the teams and players themselves, primarily in terms of shared revenues. These, altogether with the specific teams' revenues, contribute to significantly lucrative contracts for the players. For reference, the current largest NBA contract is that held by Boston Celtics' Jayson Tatum, who signed in June, 2024 a 5-year, $314 million deal with the organization.

1.3 Purpose and Content of This Final Work

To conclude the introductory section (Section 1) of the following paper, it is necessary to clearly present the purpose of our analysis, as well as provide an overview on the content presented in the paper itself.

Section 2 goes over a general introduction of basketball as a sport, as well as provides an overview of the National Basketball Association from a historical perspective (Section 2.1). Subsequently, this section provides a more in-depth analysis focused on the evolution and changes in the way basketball is played in the NBA (Section 2.2). Specifically, our analysis is centered on some major trends that highlight the evolution in playing styles: 1) Section 2.2.1 presents the evolution of pace in the NBA, starting from the 1996-97 Regular Season until the recently concluded 2023-24 Regular Season; 2) Section 2.2.2 goes over the revolution that took place in recent years thanks to the introduction of the 3-point shot.

Subsequently, through Section 3 the focus is on pursuing Exploratory Data Analysis as well as Cluster Analysis on a dataset with current NBA players' traditional and advanced statistics. Specifically, Section 3.2.1 focuses on a Principal Component Analysis that aims at identifying the grouping among observations. Building upon the results obtained, Section 3.3 consists in developing a k-means clustering model in order to identify similar groups among players. The purpose of this analysis is that of grouping players basing on their traditional and advanced statistics, without taking into account their relative position on the court. This fostered the emergence of 5 new groups of players, each with their own strengths and weaknesses.

Eventually, Section 4 builds upon the results obtained in previous sections by providing an analysis of the primary characteristics of each cluster. This allows to identify five new positions that encompass players' similarities in their statistical production. Subsequently, this section highlights relevant results and insights obtained throughout this paper. Eventually, the section also provides a suggestion on possible alternative uses of the model.

1.4 Data Collection Process

In order to properly carry out the analysis necessary for the development of our model, the appropriate data will be scraped using programming language R in the RStudio environment by relying primarily on R packages `httr`[1] and `jsonlite`[2] package.

The primary websites offering such data are:

- **NBA Official Website**: Official website of the National Basketball Association. Provides a wide range of traditional and advanced statistics related to the league itself;
- **Basketball-Reference**: Section of Sports Reference that focuses on basketball. Provides a range of statistics and data related not only to the NBA but also to other international leagues and competitions;
- **Inpredictable**: Website providing statistics and data primarily centered on the NBA and WNBA. The majority of available stats include pace and tracking data, as well as clutch, on/off and efficiency data.

[1] A tool to facilitate interaction with HTTP that allows to obtain raw data from a `url`

[2] JSON parser and generator to build data pipelines and interact with web APIs

2 NBA Introduction and Trends

Before diving into the specifics of how the game of basketball played in the National Basketball Association has evolved, it is first and foremost necessary tointroduce some background on the sport itself.

2.1 Basketball and NBA Introduction

Basketball was initially invented by Canadian-American physical educator James Naismith in 1891; at the time, Naismith's basketball rule book only contained 13 rules. For comparison, the current NBA rule book features about 65 pages of rules. The game of basketball was first featured in the 1904 Olympics, only as a demonstration sport. Subsequently, it became an official Olympic sport at the 1936 summer Olympics, in Berlin.

Building on the previous introduction of basketball, it is worth mentioning some background information on the National Basketball Association (NBA). The NBA was created in 1949 as the merger of the Basketball Association of America (BAA) and the National Basketball League (NBL); more specifically, the globally recognized founding date is 1946, as during the merger the NBL adopted the BAA's history as its own. The final step towards today's NBA took place in 1976, when the National Basketball Association merged with the American Basketball Association (ABA). As of now, the NBA consists of 30 North-American teams — one of which is actually based in Toronto, Canada, the Toronto Raptors.

2.2 NBA Gameplay Evolution

Despite the fact that basketball is played fundamentally the same way as it was initially conceived — with two opposite baskets and one ball — there have been some major milestones that have significantly contributed to the evolution of the game itself, especially in terms of improved tactics.

2.2.1 Evolution of Pace

To begin the analysis on the changes that contributed to shape the way modern basketball is played, the primary statistic that we are going to take into consideration is *pace*[3].

The introduction of the concept of pace can be traced back to the 1954-55 Regular Season: the year in which the shot clock was introduced. Up until that season, teams with possession of the ball did not have a time limit to conclude the offensive action. This in turn resulted in stagnant games, while also negatively affecting the defense's ability to recover the ball as well as fans' entertainment during the games. In light of this, Daniel Biasone (February, 1909 - May, 1992) and Leo Ferris (May, 1917 - June, 1993) — respectively, owner and general manager of the Syracuse Nationals (now the Philadelphia 76ers) — strongly lobbied for the NBA to adopt a shot clock [6]. The two are responsible for establishing the shot clock at 24 seconds, the same value it has today in both NBA and FIBA basketball. Specifically, the value was calculated by Ferris by dividing the number of seconds in a game (2,880) by the average number of shots taken in a game (60) — thus the 24-second limit per possession.

The introduction of the shot clock immediately increased the pace at which games were played: during the 1954-55 season, average scoring per game increased to 93.1 points, up from the 79.5 average of the previous season — a 17.1% increase. It therefore appears that the introduction of the shot clock caused the game to be played at a higher pace. This indeed represents the first evident trend of the last two decades: games are played at a higher speed, with a constant increase in the number of offensive possessions played by each team over 48 minutes.

It is possible to clearly represent the evolution in pace trends over the years by referencing the data that is publicly available on the NBA website; more specifically, in this case the primary sources of data are the advanced stats — expressed on a per-game basis — collected for each of the 30 NBA teams during the Regular Season. Eventually, the following analysis will be taking into consideration data available from the 1996-97 season,

[3]Defined as the number of offensive possession per 48 minutes for a team or a player [11]

the first year tracking data is available, up until the recently concluded 2023-24 season.

After having identified the appropriate data of interest, it is necessary to use programming language R in the RStudio environment to write a short script that allows to scrape data directly from the NBA website. Such script allows to obtain a single data frame containing 29 or 30 rows — depending on the number of NBA teams during that specific season — and 46 columns, each containing teams' advanced stats gathered during the specific NBA Regular Season, and expressed on a per-game basis. The resulting data frame looks like this:

NBA Advanced Stats, 2023-24 Regular Season							
`TEAM_ID`	`TEAM_NAME`	`GP`	`W`	`L`	...	`PACE`	`Season`
1610612737	Atlanta Hawks	82	36	46	...	100.84	2023-24
1610612738	Boston Celtics	82	64	18	...	97.98	2023-24
1610612751	Brooklyn Nets	82	32	50	...	97.56	2023-24
1610612766	Charlotte Hornets	82	21	61	...	97.82	2023-24
1610612741	Chicago Bulls	82	39	43	...	96.94	2023-24
1610612739	Cleveland Cavaliers	82	48	34	...	97.62	2023-24

Table 1: Sample of teams' pace statistics, 2023-24 Regular Season

As represented by Table 1, the data frame lists every NBA team during the specific Regular Season, altogether with some relevant stats:

1. The individual identification code (`ID`) for each team;
2. The name of each team (`TEAM_NAME`);
3. The number of games played (`GP`), which are normally 82;
4. The number of wins (`W`);
5. The number of losses (`L`);
6. The pace (`PACE`), an estimate for the number of possessions per a 48-minute game for a team.

Nonetheless, the results obtained above are not satisfactory: as the primary interest is on the trend of pace since the 1996-97 Regular Season, it is necessary to obtain data for all the other seasons. Rather than doing it manually by altering the script each time to adapt it to the season of interest, it is sufficient to modify the script into a simple function that iterates scraping for each of the seasons of interest. Eventually, all the individual data frames are merged into one — now containing 832 rows of data – so that each sequence of 29 or 30 rows (depending on the number of teams being part of the NBA during that Season) contains data for each team during that specific Regular Season.

The last step in the analysis of the evolution of pace is to reduce the previously obtained data frame to a simpler one composed by two columns:

`Season`	`avg_pace`
1996-97	91.6
1997-98	91.8
1998-99	90.5

Table 2: Sample of yearly NBA average pace

In Table 2 the Season (`Season`) column indicates the NBA Regular Season to which the data refers; the Average Pace (`avg_pace`) column indicates the average pace of all NBA teams during the specific season.

The data is ready to be plotted using R's `ggplot2` package. The resulting scatter plot, presented in Fig. 1, highlights an overall stable trend in average pace league wide, until the 2012 NBA Regular Season. Indeed, after 2012 — when average league pace was 92.3 — average pace increases almost every year, reaching its peak at 100.8, during the 2020 season — the season of the NBA Bubble in Orlando. During the recently concluded 2023-24 season, average pace was 99.2. Such value corresponds to an 8.42% increment if compared to 20 years ago, during the 2003-04 season.

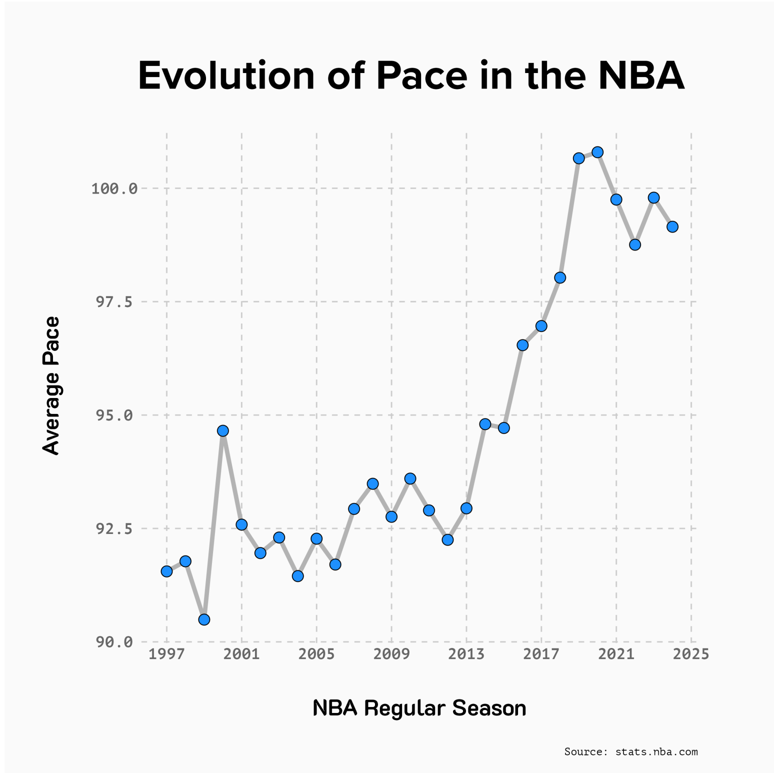

Figure 1: Progression of NBA average pace since 1996-97

Building on top of this, to further investigate the revolution in the way basketball is played in terms of speed, it is possible to introduce an alternative measure of playing speed other than pace: seconds per possession.

By referencing data available on the **inpredictable** website, it is possible to pursue an analysis similar to that outlined before, in order to obtain a visualization representing the progression in possession length over the years. The first step in this process is to scrape historical possession data tracing back to the 1996-97 Regular Season directly from the **inpredictable** website, in order to obtain a data frame whose first few rows look like this:

Teams' Possession Stats					
Team	Gms	poss	seconds_per	league_rank	Season
PHX	82	7,865	14.6	1	1997
PHI	82	8,101	14.7	2	1997
WAS	82	7,817	14.8	3	1997
BOS	82	8,104	14.8	4	1997
SEA	80	7,410	14.9	5	1997

Table 3: Sample of teams' possession data for each Regular Season

Now it is sufficient to create a new data frame that groups the data contained in Table 3 in two distinct columns: Season (`Season`), which indicates the NBA Regular Season to which the data refers, and Average Seconds (`avg_seconds`), which indicates the average seconds per possession of all NBA teams during the specific season.

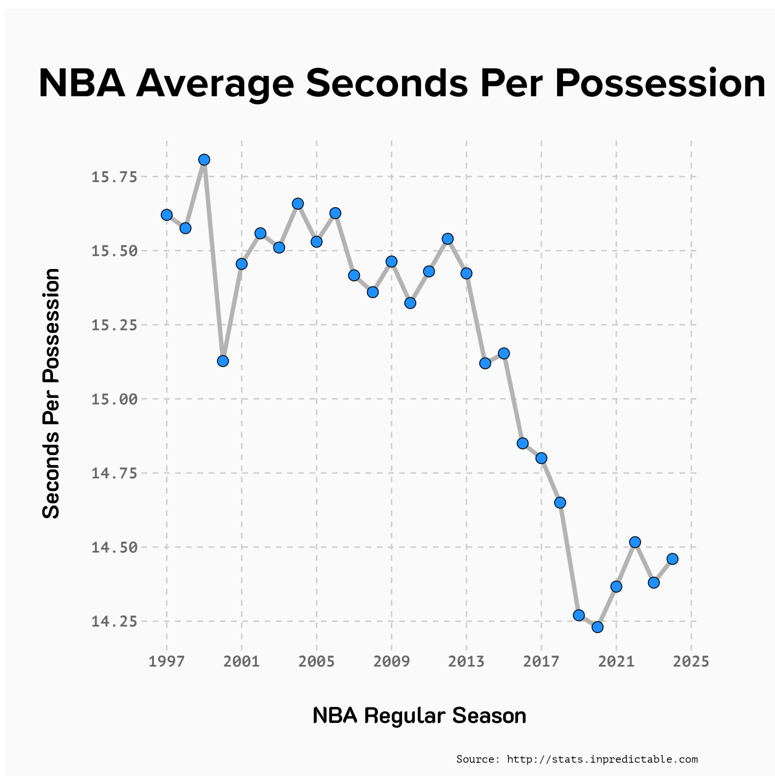

Figure 2: Progression of NBA average seconds per possession

Taking into consideration both Fig. 1 and Fig. 2, it appears clear that since the 1996-

97 NBA Regular Season, there has been a clear correlation between a team's pace during a game and the number of seconds used per each offensive possession. Indeed, the shift towards a speedier style of play is evidenced by the gradual decrease in the average time a team holds onto the ball during an offensive possession. Specifically, the new offensive philosophy starts spreading with the new millennium – despite initially being measured in approach. Subsequently, this philosophy starts taking significant hold since the beginning of the 2010s. Indeed, since the year 2012, it is possible to clearly identify a steady *decline* in average possession length; on the other hand, in Fig. 1 the year 2012 corresponds to the beginning of a steady *increase* in league average pace. Teams are now eager to capitalize on a greater number of scoring opportunities thanks to a speedier style of play: quicker possessions translate to an increased volume of offensive possessions – and thus scoring opportunities – during a game.

Such correlation between the average seconds used per offensive possession and the number of Field Goal Attempts[4] taken during a game can be visualized through the following scatter plot, altogether with a regression line:

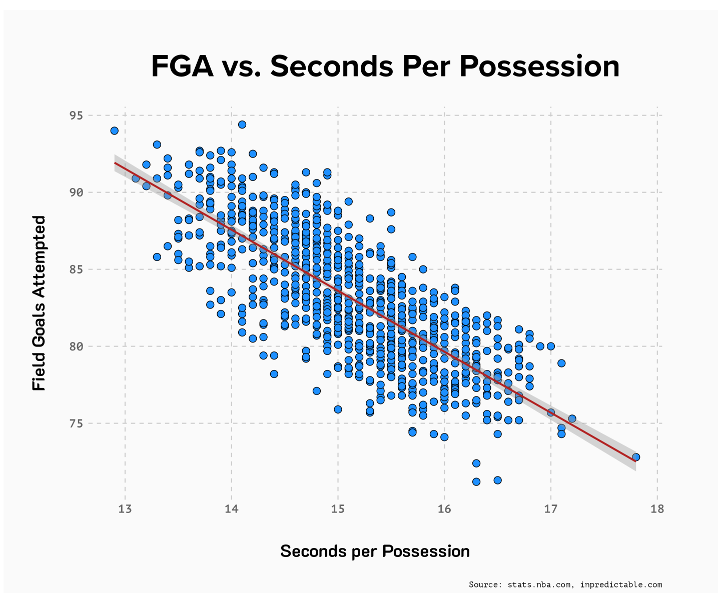

Figure 3: Correlation between Field Goals Attempted and seconds per possession

[4]The number of Field Goals (shots) taken by a player or a team. Include both 2-pointers and 3-pointers

From Fig. 3 it is possible to conclude that as teams opt for a quicker playing style, thus reducing the average duration of each possession, this results in a higher volume of scoring opportunities. Therefore, shorter possessions – in terms of seconds used – normally result in more scoring opportunities (Field Goal Attempts) taken by a team, and thus also in more points scored during a game.

Moreover, thanks to R's `lm()`[5] function, it is possible to derive a linear model for the correlation between Field Goal Attempts (`FGA`) and Seconds per Possession (`seconds_per`). Using the data from Fig. 3, it is possible to derive a linear model of the type:

$$\texttt{FGA} = 143.03 - 3.962 * \texttt{seconds\_per}$$

Specifically, the statistically-significant model has multiple R-squared $R^2 = 0.5805$, implying that 58.05% of the variation in the dependent variable `FGA` can be explained by changes in the independent variable `seconds_per`. Ultimately, the coefficient estimate of `seconds_per` (-3.962) implies that each additional one-unit increase in `seconds_per` is associated with an average *decrease* of 3.962 in `FGA`.

To conclude this section on the evolution of pace and the subsequent implications for the game of basketball, it is necessary to assess what are the major statistical contributors to a team's winning percentage (`W%`). Using R's `corrplot()`[6] function it is possible to derive a correlation matrix between a team's traditional statistics:

[5] Function used to fit linear models to a data frame in R programming language

[6] Part of the `corrplot` package, which provides a visual exploratory tool on correlation matrices

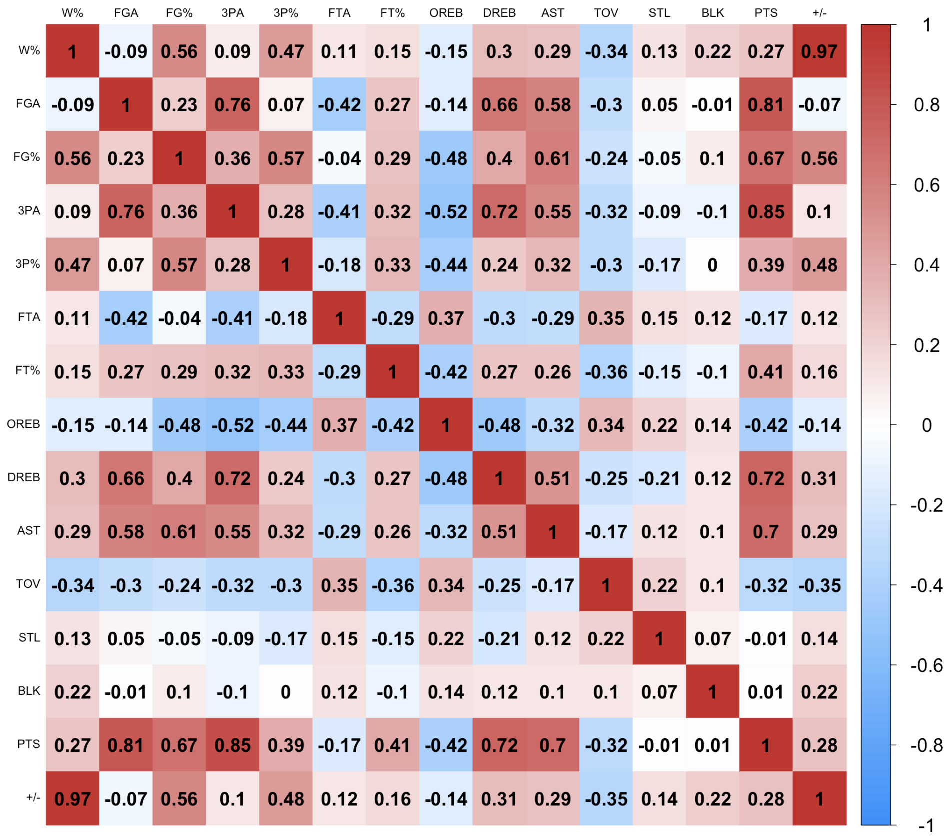

Figure 4: Correlation matrix between teams' traditional stats

The correlation matrix in Fig. 4 highlights that a team's winning percentage (`W%`) is primarily influenced by their plus-minus[7] (`+/-`). Other factors that influence winning percentage are statistics related to a team's offensive output, primarily points scored (`PTS`) and field goal percentages (`FG%` and `3P%`). Fig. 4 also emphasizes a positive correlation between winning percentage and defensive rebounds (`DREB` – more shots missed and less second chance opportunities for the opposing team), altogether with assists (`AST`). Conversely, the matrix also highlights a negative correlation of turnovers (`TOV`) to winning, thus highlighting the importance of maintaining possession and control of the ball when on offense.

Having identified the major statistical contributors to winning percentage, it is now possible to explore the correlation between such factors and a team's pace. Specifically,

[7] Plus-minus is the point differential when a player is on the floor versus when the player is not on the floor

this step of the analysis will take into consideration teams' rankings in every statistical category, within the NBA (*i.e.:* a rank of 1 implies that the team is leading the Association in that category; conversely, a rank of 30 implies that the team places last out of 30 teams in that category). The purpose of this analysis is to assess whether teams that play at a faster pace also tend to have higher rankings in the statistical categories that majorly contribute to winning percentage:

Figure 5: Correlation matrix between teams' rankings in traditional stats

In light of Fig. 5, it is possible to conclude that a team's speed of play has a direct and positive correlation with scoring and assisting (`PTS_Rank` and `AST_Rank`), thus reinforcing the previous theory that a faster pace leads to more points scored. On the other hand, pace has a negative correlation with turnovers (`TOV_Rank`), thus emphasizing the trade-off between a faster offense and ball security. Ultimately, pace has a weak correlation with shooting percentages: the correlation is slightly positive with `FG%_Rank` and slightly negative with `3P%_Rank`, thus suggesting that swift play style does not necessarily affect shooting percentages.

To conclude, the evolution of pace within the NBA represents a broad evolution within the game itself, rather than being directly correlated to team success. Moreover, this has had a significant impact on players themselves: the increase in playing speed fostered the emergence of smaller and faster guards that now tend to have a major offensive advantage. This is in contrast with the way basketball was played decades ago, with centers typically dominating – especially in the restricted area – thanks to a significant size advantage. Such shift towards a speedier style of play was also fostered by the increase in the use of the 3-point shot, whose effects will be analyzed in the following Section 2.2.2.

Overall, the pace of a team's offense is not a direct indicator for success, but rather is an outcome of a broader shift towards a more dynamic style of play that primarily relies on increased spacing and speed.

2.2.2 Three-point Revolution

The second, major revolution that contributed to the emergence of modern basketball is the introduction of the three-point field goal – a shot that is taken behind the three-point line, whose distance from the basket is normally 7.24 meters, and 6.70 meters in the corners.

The three-point field goal was first introduced by the American Basketball Association starting from the 1967-68 season, as the league commissioner believed that this new kind of field goal would give smaller players an advantage and an additional chance to score, thus also spacing the offense and forcing the defense to open up more. The field goal was then introduced in the National Basketball Association during the 1979-80 NBA Regular Season. This contributed to significantly altering game planning and tactics, mainly because of the new offensive spacing which is generated by relying more on the three-point shot.

By referencing the traditional teams' statistics available on the NBA website, it is possible to proceed with an analysis on the impact that the introduction of the three-point field goal has had on shaping the way modern basketball is played.

The first dataset that will be used contains traditional statistics for each team starting from the 1996-97 NBA Regular Season, thus making for 832 rows of statistics, expressed on a per-game basis:

Teams' Traditional Stats							
Team_ID	Team_NAME	GP	...	FG3M	FG3A	FG3_PCT	Season
1610612737	Atlanta Hawks	82	...	8.0	22.4	0.360	1996-97
1610612738	Boston Celtics	82	...	5.7	16.2	0.351	1996-97
1610612766	Charlotte Hornets	82	...	7.2	16.9	0.428	1996-97
1610612741	Chicago Bulls	82	...	6.4	17.1	0.373	1996-97
1610612739	Cleveland Cavaliers	82	...	5.9	15.7	0.376	1996-97

Table 4: Sample of teams' traditional statistics for each Regular Season

Starting from the dataset reported in Table 4, it is possible summarise it and derive a data frame that reports the average number of three-pointers both made and attempted during a specific Regular Season across the whole National Basketball Association:

three_avg	pct_avg	year
16.8	0.360	1997
12.7	0.344	1998
13.2	0.336	1999
13.7	0.353	2000

Table 5: Data frame summarising league-wide 3-point statistics

Eventually, starting from Table 5 it is possible to derive a scatter plot displaying the average number of three-point field goals attempted within the National Basketball Association for each Regular Season, starting from 1996-97.

Despite an initial learning curve, it appears evident from Fig. 6 that the use of the three-point shot constantly increased since its introduction: during the 1997-98 NBA Regular Season, the average number of three-point attempts taken during a game league-wide was only 12.7; the corresponding number for the recently concluded 2023-24 Regular Season is instead 35.1. This indicates that in the span of two decades, the average number of three-point field goals attempted during a game has increased 176%.

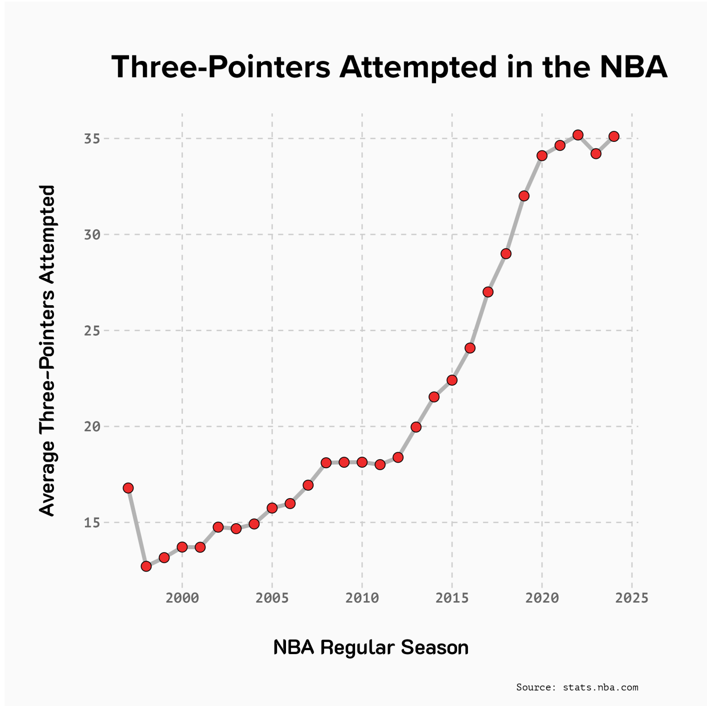

Figure 6: Progression of average 3-pointers attempted per game in the NBA

Indeed, the increased reliance on the three-point shot – especially thanks to its contribution to a more fast-paced and fluent offense – strengthened the adoption of a three-point field goal if compared to other field goals, that despite being "easier" to make also translate to less expected points.

To visualize such trend we will rely on another data frame containing information on the average total number of shots taken league-wide for each individual Regular Season. Specifically, the dataset scraped directly from the NBA website appears like the following:

Season	avg_FGM	avg_FGA	avg_FG3M	avg_FG3A	pct
1996-97	2958	6503	496	1377	0.212
1997-98	2944	6536	360	1043	0.160
1998-99	1709	3910	223	658	0.168
1999-00	3020	6732	397	1125	0.167

Table 6: Data frame summarising average total shots for each Regular Season

Where:

1. `Season` indicates the Regular Season to which the data refers to;
2. `avg_FGM` indicates the average number of field goals made by a team;
3. `avg_FGA` indicates the average number of field goals attempted by a team;
4. `avg_FG3M` indicates the average number of three-point field goals made by a team;
5. `avg_FG3A` indicates the average number of three-point field goals attempted by a team;
6. `pct` indicates the number of three-point field goals attempted, expressed as a percentage of total field goals attempted.

By referencing the data available from Table 6, it is possible to derive a bar chart to highlight the progression of the league-average total number of three-point field goals attempted, expressed as a percentage of the total number of field goals attempted. The trend is represented in Fig. 7:

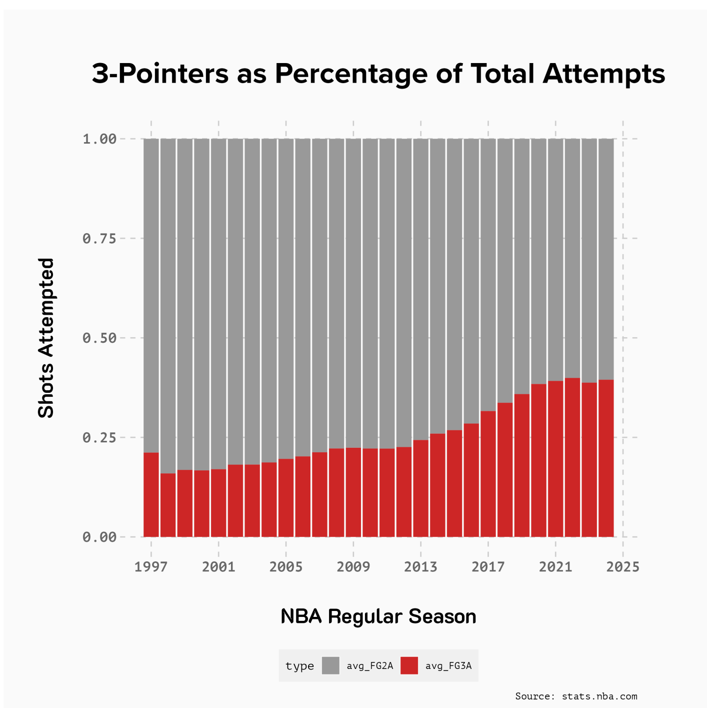

Figure 7: Progression of 3-point shots as a percentage of total field goals attempted

From Fig. 7 it is possible to highlight the constant increase in the number of three-point field goals attempted, expressed as a percentage of the total shots attempted during a game. From the graph it appears evident that NBA teams rely on three-pointers for a major portion of their offensive shots taken. Fig. 7 is also particularly useful in highlighting the progression of such trend: only a decade ago, during the 2012-13 NBA Regular Season, the number of three-pointers attempted as a percentage of total shots was slightly lower than 25%. For the 2023-24 Regular Season this number is instead 39.45% – only slightly lower than the peak of the 2021-22 Regular Season, when a staggering 39.92% of the total field goals attempted were from three-point range.

To conclude the overview on the revolution deriving from the introduction of the three-point shot, it is possible to compare the evolution of the distribution of the total number of three-point field goals attempted by each team during a Regular Season. By comparing the different box-plots obtained for each Regular Season, it is possible to grasp important insights on the evolution of the distribution of three-point attempts. Specifically, we will once again refer to a data frame containing traditional statistics – as Fig. 6 – that will instead be expressed as totals for the whole Regular Season, rather than being expressed on a per-game basis.

Starting from this dataset, by relying on R's **geom_boxplot()**[8] function, it is possible to obtain a graph that displays the distribution of the total number of three-point shots attempted league-wide during a specific Regular Season. Box-plots are particularly relevant as they allow to visualize the distribution of a continuous variable along five major statistics: the minimum, the first quartile, the median, the third quartile and the maximum. Additionally, box-plots also allow to plot outliers in the distribution.

The graph presented in Fig. 8 allows to quickly visualize the differences in the distribution of total three-pointers attempted year after year. Slightly less than two decades ago, during the 2004-05 NBA Regular Season, the median number of total three-point field goals taken during the season was 1,266. During the recently concluded 2023-24 Regular Season, the median number of total three-pointers attempted was instead 2,848

[8] `geom_boxplot()` allows to plot the box-plot of a continuous variable to display its distribution along five summary statistics

– a 125% increase.

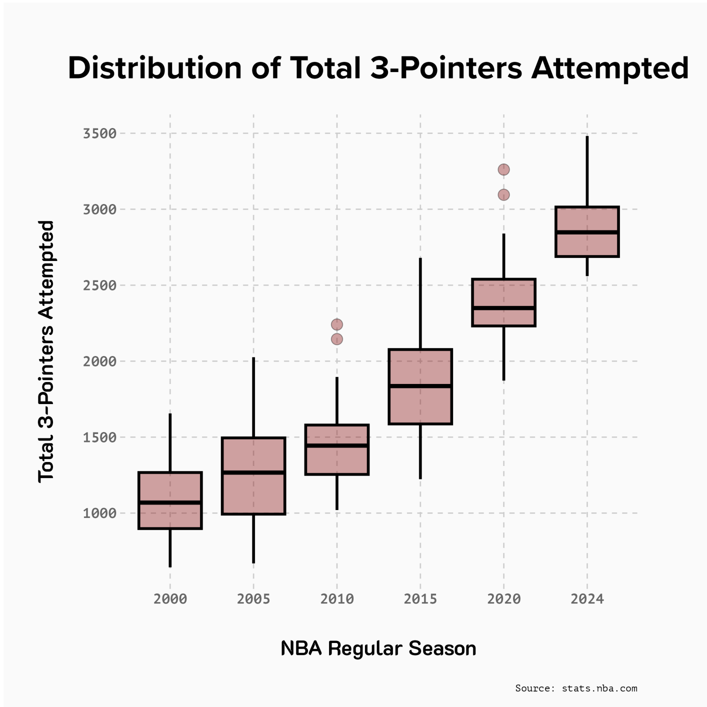

Figure 8: Distribution of average total three-point shots attempted yearly

Moreover, Fig. 8 highlights the presence of two outliers for the 2009-10 Regular Season: the New York Knicks and the Orlando Magic, respectively with 2,145 and 2,241 total three-pointers attempted. Such numbers are significantly lower than the mean number of three-pointers attempted during the 2023-24 Regular Season, and would place in the first quartile for this same season. This therefore highlights the significant shift in play-style and the constantly-increasing reliance of teams on the use of the three-point shot.

To conclude the following section, it is necessary to assess how three-pointers correlate to team success. By referencing Fig. 4, which displays a correlation matrix between teams' traditional statistics, it is possible to derive that the effectiveness of a team in shooting from three – measured as three-point field goal percentage – is a major factor that influences winning percentage of a team (the correlation factor between the two is 0.47). Such analysis is also strengthened by Fig. 9, a scatter plot displaying the win percentage correlation with the shooting percentage from three-point range. The graph highlights a positive but weak correlation between the two variables, and thanks to R's

lm() function it is possible to derive a linear model of the type:

$$\texttt{W\_PCT} = 0.441 + 0.0076 * \texttt{FG3M}$$

Here `W_PCT` indicates a team's winning percentage and `FG3M` indicates the number of three-pointers made during a game. Specifically, the model has a multiple R-squared $R^2 = 0.024$, implying that only 2.4% of variation in the dependent variable `W_PCT` can be explained by changes in the independent variable `FG3M`. Moreover, the coefficient estimate of `FG3M` (+ 0.0076) indicates that each additional one-unit increase in `FG3M` is associated with an average increase of 0.0076 in `W_PCT`.

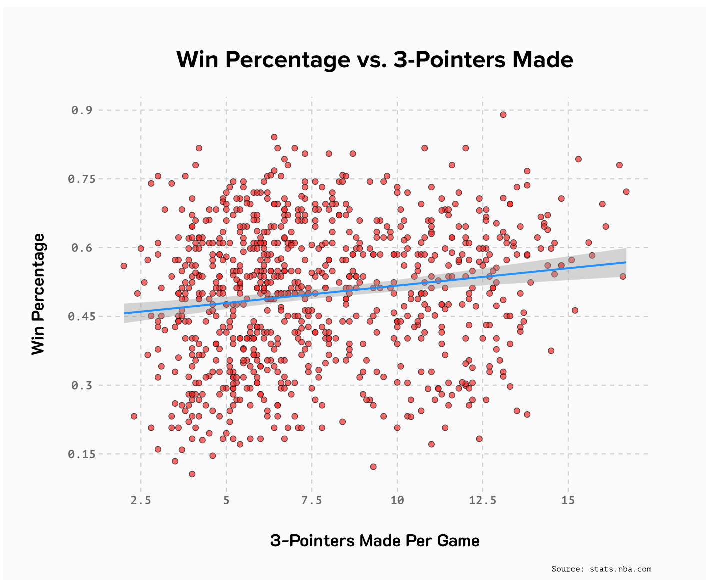

Figure 9: Correlation between win percentage and three-point shooting percentage

From Fig. 9 follows that the relationship between the number of three-point field goals made during a game (`FG3M`) and the winning percentage of a team during a Regular Season (`W_PCT`) is a positive but weak one: indeed, to a one-unit increase in the number of three-pointers made only corresponds a slight increase in the winning percentage of a team.

Despite such correlation, it appears evident that the introduction of the three-point field goal significantly altered the pace at which the game of basketball is played: the reliance on the three-pointer allows teams to space more their offense, thus leading to a more fluent playing style that empowers players to make more rapid decisions and exploiting the errors made by the defensive team.

Having thus concluded the analysis on the two major trends that contributed to altering the style at which professional basketball is played, it is eventually of interest to assess the repercussions of such changes on teams and their rosters.

3 Clustering Model

In Section 2.2.1 the primary point of focus was on assessing the evolution in pace that took place within the National Basketball Association in the last two decades. Additionally, we assessed the implications that a faster pace have on team winning: the evolution of pace is not directly correlated to team success, but has contributed to significantly shape the playing styles adopted by teams as well as players.

In Section 2.2.2 instead, the primary point of focus was on the revolution that took place since players and teams started relying more and more constantly on the use of the three-point field goal. Despite its use being limited once it was first introduced, in the last decade teams as well as players have shaped their offensive game-planning strategies to rely significantly on the three-point shot, primarily due to its higher expected value if compared to other shots, but also thanks to its other offensive implications: relying on perimeter shooting allows teams to be more spaced, thus being able to generate more scoring opportunities and putting a significant strain on the defensive team.

In light of what is presented above, it appears clear that the archetype of the ideal basketball player has drastically changed in the last two decades. Therefore, the following section of this paper will focus on developing a clustering model that groups players basing on their similarities, rather than relying on their traditionally-assigned positions.

Before diving into the development of a clustering model, it is first necessary to provide an overview of the classical players' positions on the court [8], which are the following:

1. **Point Guard**. Players in this position normally set the team's offensive and defensive pace. This is due to their ability to generate offense by analyzing the game; additionally, they often possess outstanding passing skills. Thus, they normally bring the ball up the floor and are responsible for quick decision-making;

2. **Shooting Guard**. This archetype of player shines at shooting the ball, especially from three. Additionally, they also possess elite touch, thus being able to score in other scenarios, thriving particularly in catch and shoot scenarios;

3. **Small Forward**. These normally are flexible players that contribute significantly to rebounding, but are also viable offensive options. They may be responsible for ball handling. Additionally, they normally occupy the space on the floor comprehended between the rim and the three-point line.

4. **Power Forward**. These players play primarily around the area, both on offense and on defense. They also excel at rebounding and rim protection, particularly thanks to their size and physical strength;

5. **Center**. The key responsibilities of players in this role is that of rebounding, blocking shots and finishing around the rim. Indeed, centers are often tall players that in turn lack agility. This makes them particularly efficient near the basket.

3.1 Data Collection

Before diving into the development of a clustering model, it is first and foremost necessary to collect all the necessary data, which will then be cleaned using R in RStudio.

Specifically, for this section we will rely on three major datasets containing information and statistics on NBA players for the recently concluded 2023-24 NBA Regular Season.

The primary data source that will be taken into consideration is the **National Basketball Association website**. It will be a source for players' traditional statistics, expressed as Regular Season totals. Thanks to the use of R's `httr` and `jsonlite` packages, it is possible to obtain a final, clean dataset that looks like the following:

`Player`	`...`	`G`	`MP`	`FG`	`FGA`	`X3P`	`...`
A.J. Lawson	...	42	310.9	54	121	13	...
Aaron Gordon	...	73	2296.8	398	716	40	...
Aaron Holiday	...	78	1269.3	186	417	84	...
Aaron Nesmith	...	72	1994.6	315	635	140	...

Table 7: Sample of players' total traditional statistics, 2023-24 Regular Season

Table 7 is limited to the first four rows and six columns, but the whole data frame that will be used for the analysis contains observations for 541 players' traditional, total statistics for the 2023-24 Regular Season. Such traditional statistics include the number of total: games played (`G`), minutes played (`MP`), field goals made (`FG`), field goals attempted (`FGA`), three-point field goals made (`X3P`), three-point field goals attempted (`X3PA`), free-throws made (`FT`), free-throws attempted (`FTA`), total rebounds (`TRB`), assists (`AST`) steals (`STL`), blocks (`BLK`) and points (`PTS`).

The data frame in Table 7 reports only the total statistics for the whole 2023-24 NBA Regular Season, but thanks to the information it contains, it is possible to create additional columns that contain traditional statistics expressed on a per-game basis. It is sufficient to divide the specific, total statistic of interest by the number of games played by the specific player. This allows to obtain not only total traditional statistics, but also per-game traditional statistics for 541 players.

Moving on with the second data collection process, it is now necessary to reference the advanced statistics that are available on the **BasketballReference** website. In this case, thanks to the use of the `rvest` package it is possible to derive the following data frame:

Player	Pos	PER	BPM	VORP
A.J. Green	SG	10.5	-2.1	0.0
A.J. Lawson	SG	11.2	-4.6	-0.2
AJ Griffin	SF	1.2	-9.6	-0.3
Aaron Gordon	PF	16.8	1.3	1.9

Table 8: Sample of players' advanced statistics, 2023-24 Regular Season

Specifically, the data frame summarized in Table 8 reports the player name (`Player`), the player position (`Pos`) and three advanced statistics:

1. `PER`: the Player Efficiency Rating [2] summarizes a player's statistics and accomplishments into one single value that is expressed on a per-minute basis. It is not an all-in-one statistic, but it allows to compare players' per-minute productivity while taking into account other major statistics. League average is 15.00 every year;

2. `BPM`: the Box Plus/Minus [7] is a metric that takes into account a player's traditional statistics (thus, not using play-by-play data) to estimate the player's contribution to the team when that player is on the court. The player's contribution to the team is measured in points above league average per 100 possessions. In this case, league average is defined as 0.0;

3. `VORP`: the Value over Replacement Player [7] is a metric that converts the Box PlusMinus (BPM) into an estimate of the player's contribution to the team, measured against what a "replacement player[9]" would provide. The `VORP` value for a replacement player is -2.0 in the NBA.

The contents of both data frames will be merged into one comprehensive data frame which thus contains: players' total traditional statistics, players' per-game traditional statistics and players' major advanced statistics (`PER`, `BPM`, `VORP`) for the 2023-24 NBA Regular Season.

[9] A replacement player is defined as a player on a minimum salary or not a member of a teamś rotation

3.2 Exploratory Data Analysis

Having now obtained a single data frame containing all relevant statistics, it is possible to move on to the penultimate step before beginning with clustering. Such section will be dedicated to exploratory data analysis techniques on the previously-obtained data frame, or on parts of it.

We start by focusing primarily on traditional statistics, expressed on a per-game basis, and therefore on the data frame of Table 7. In this case we are dealing with 541 observations for a total of 14 per-game statistics, which include but are not limited to: points per game (`PTS_pg`), total rebounds per game (`TRB_pg`), assists per game (`AST_pg`), minutes played per game (`MP_pg`) and field goals made per game (`FG_pg`). It follows that such observations are gathered on several different scales: points per game range from 0 to almost 35, whereas steals per game range from 0 to 2.5. Therefore, in order to assess the spread of each statistic, it is possible to use R's `var()` function in order to obtain a measure of the variance of each per-game statistic:

	Variance
`MP_pg`	98.40
`FG_pg`	5.96
...	...
`BLK_pg`	0.17
`PTS_pg`	45.43

Table 9: Sample of per-game statistics' variance

In light of the results expressed in Table 9 on the variance of the per-game statistics, it is possible to conclude the following: given the discrepancy in spread between the different statistics considered, it is necessary to scale the data frame. Indeed, scaling each of the per-game stats ensures that no single feature dominates subsequent analyses only due to the method used to measure the statistic. Scaling the data frame can be achieved by relying on base R's `scale()`[10] function, which allows to standardize the values of the per-game statistics. It follows that, after the standardization, each per-game stat column will have a mean equal to 0 and a standard deviation equal to 1.

3.2.1 Principal Component Analysis

Having now standardized the per-game statistics for each player, it is possible to move on to another crucial step before clustering our data. The following step is that of Principal Component Analysis, whose purpose is that of determining whether there are any inherent groupings among players.

Principal Component Analysis (PCA) is a statistical technique whose primary application is that of reducing a dataset by grouping the information contained in its variables into a smaller dataset, while maintaining the most important information of the dataset. Specifically, Principal Component Analysis summarizes the information contained in the original variables into a new set of uncorrelated variables – the *principal components* – so that the average distance between the observations in the dataset is minimized. It follows that principal components are normally arranged basing on the extent to which they can explain variance within the dataset. Before performing Principal Component Analysis it is necessary that the variables within the dataset of interest are centered and normalized. Once PCA has been carried out, the number of components should equal the number of variables.

In order to assess the importance of each component, it is sufficient to apply R's `prcomp()`[11] function to our dataset containing the standardized, per-game statistics. The result is summarized in the following table:

[10] `scale()` allows to center and/or scale the columns of a numeric matrix

[11] `prcomp()` performs a Principal Component Analysis on the the given data set

Importance of Components							
Stat	**PC1**	**PC2**	**PC3**	**PC4**	**PC5**	**PC6**	**PC7**
Standard deviation	3.17	1.32	0.91	0.73	0.54	0.51	0.43
Proportion of Variance	0.72	0.12	0.06	0.04	0.02	0.02	0.01
Cumulative Proportion	0.72	0.84	0.90124	0.94	0.96	0.97	0.99202

Table 10: Principal Component summary table

From Table 10 it follows that the first two principal components (**PC1** and **PC2**) explain a significant majority of the variance within the dataset. Consequently, we will focus on these two features when assessing similarities between players. Specifically, it is possible to plot the principal components obtained from the PCA by relying on R's `autoplot()`[12] function. The following plot will take into consideration the first two principal components, given the extent to which they explain the variance within the dataset:

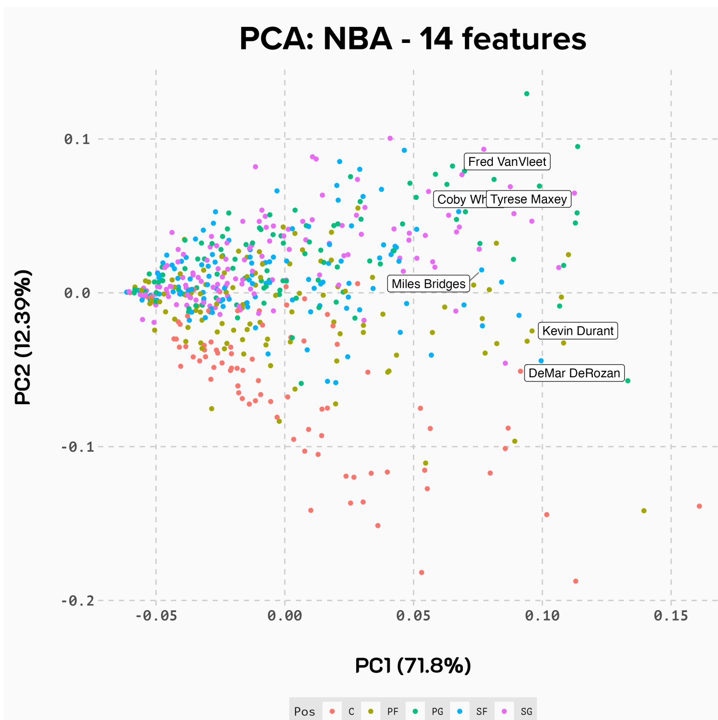

Figure 10: Plot of primary Principal Components explaining variance within the dataset

[12]`autoplot()` allows to generate different types of plots depending on the object passed to it.

By examining Fig. 10 it appears evident that there are major differences between players based on their overall statistics, as well as similarities based on their positions. For instance, it seems that the majority of the so-called "All-Stars[13]" may be grouped together. Additionally, Centers and Power Forwards are more present in the bottom of the chart, whereas Shooting Guards are more condensed in the upper portion of the chart.

3.3 Cluster Analysis

After an analysis on the variance of measurements in our dataset, it is now time to move onto the conclusive part of this paper. The purpose of this section is that of developing a clustering model that allows to group players according to a series of traditional and advanced statistics, in order to identify concentrations of players that share common characteristics among themselves.

Cluster Analysis refers to a technique that aims at dividing a series of observations into natural groups, known as *clusters*, according to a specific criterion. The cases in a cluster are very similar among themselves, according to the criterion, but are also very different from the ones contained in other clusters. This technique can be particularly helpful when searching for hidden relationships within data.

The clustering method that will be used on our dataset is k-means clustering. Specifically, k-means clustering is a partitioning method which divides observations into a series of k clusters, depending on their distance from the centers of the clusters. Normally, the process of k-means clustering consists of an iteration of the following steps:

1. Identify the number of k initial cluster centers;
2. Assign each observation to the nearest cluster basing on the distance of the observation from that cluster's center;
3. Identify the centroids of the clusters[14] that were previously created;

[13] NBA players that take place in the annual All-Star Game

[14] Centroids are the geometric centers of the cluster

4. Calculate the distance of each observation from the centroids, and re-assign each observation to the nearest cluster.

To summarise, k-means clustering is an iterative technique that allows to identify natural groups within the data according to a specific criterion. The first step in pursuing this method is that of identifying the appropriate number of cluster centers in which to divide the observations. In order to identify the appropriate number of cluster centers, and therefore of clusters, it is possible to rely on the use of R's `NbClust()`[15] function, applied on the dataset containing the standardized per-game statistics. The `NbClust()` function also allows to specify the index method. In this case, the silhouette value was used.

The silhouette value measures how similar an observation is to its own cluster, against the other clusters. Normally, the value ranges from -1 to +1: a high value indicates that the observation fits with members of its own cluster and is significantly different from those of other clusters. The `NbClust` package also allows to plot the resulting graph, in order to properly identify the number of clusters:

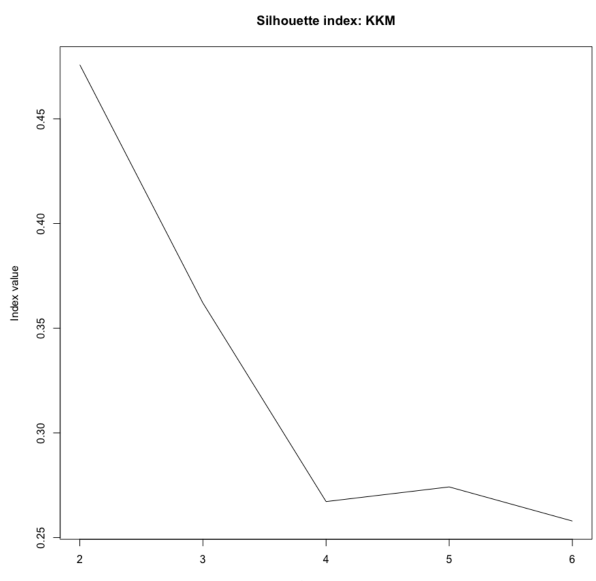

Figure 11: Plot of the silhouette value in cluster analysis

[15]The `NbClust` package provides 30 indices for determining the optimal number of clusters; it proposes the best clustering scheme basing on the different results

In light of Fig. 11, it follows that the starting number of k cluster centers that is chosen is 5. Indeed, despite a k equal to 3 results in a higher silhouette value, by running the model it would follow that observations are not optimally distributed, with several instances of players overlapping their cluster. Setting k equal to 5 allows for a clean classification of the observations in groups as different as possible from one another.

After having identified the optimal number of clusters, it is possible to proceed with the clustering analysis. The dataset of reference is the one containing players' traditional, standardized per-game statistics. Thanks to the `kmeans()`[16] function of R's `stats` package, it is possible to partition the standardized statistics into groups, such that the distance of such observations from the centers of the clusters are minimized. Subsequently, each observation is assigned a number depending on the cluster to which it belongs, so that the resulting data frame looks like the following:

Data frame with clusters							
Year	Player	Tm	km_cluster	Pos	Age	G	...
2024	A.J. Lawson	DAL	1	SG	23	42	...
2024	Aaron Gordon	DEN	4	PF	28	73	...
2024	Aaron Holiday	HOU	2	PG	27	78	...
2024	Aaron Nesmith	IND	3	SF	24	72	...

Table 11: Sample of resulting data frame with cluster information

Specifically, in Table 11 only provides the first few rows of the resulting data frame. The columns that are now represented in the Table are the ones containing players' traditional total statistics, traditional per-game statistics and advanced statistics. The primary addition to the data frame is represented by the `km_cluster` column: it indicates the cluster to which the specific player belongs. Specifically, as clusters are assigned on the basis of the statistics that were taken into consideration, it follows that lower-level clusters will group players that normally under-perform in such statistical categories. Conversely, higher-level clusters will group players that perform better in the considered statistical categories. Having now obtained the necessary information, and having assigned players

[16]Performs k-means calculations on a given data matrix.

to their corresponding clusters, it is possible to plot the observations while also grouping in clusters:

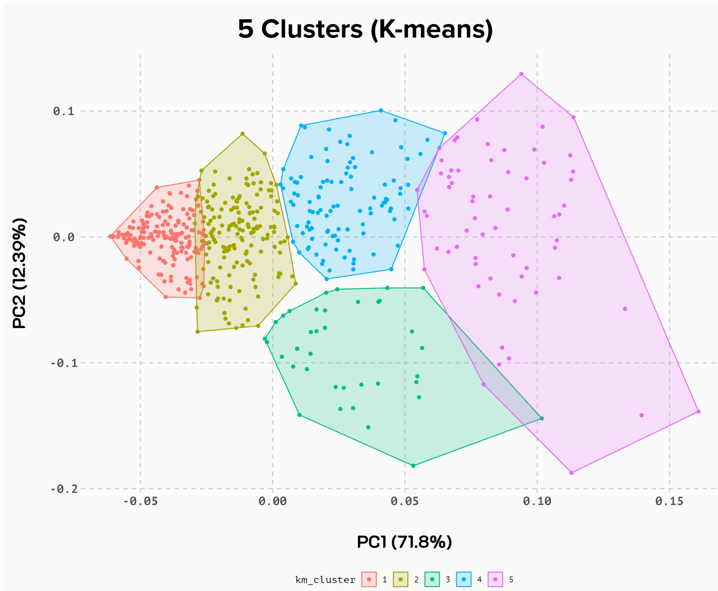

Figure 12: Plot of the observations grouped in clusters

Fig. 12 allows to display the observations along the two primary components while also grouping them into the appropriate clusters. It is possible to derive that lower-level clusters can be found in the left portion of the graph, corresponding to below-average positioning in the relevant statistical categories. Conversely, the right portion of the plot aggregates players which tend to perform at high levels, as evidenced by their statistics.

Subsequently, it is possible to obtain a data frame that summarizes the number of players belonging to each cluster, altogether with the average statistics for each cluster:

km_cluster	Count	Age	PTS_pg	...	PER	BPM
1	173	25.4	2.4	...	9.4	-4.4
2	164	26.0	6.1	...	11.9	-1.9
3	104	27.1	12.3	...	13.4	-0.7
4	37	25.1	11.7	...	18.4	0.8
5	63	27.3	22.6	...	19.5	2.6

Table 12: Data frame with cluster information and statistics

Table 12 only represents a sample of the full dataset. By examining the dataset in its entirety, it is possible to conclude that there exists an increasing hierarchy within clusters which corresponds to their statistical production. Indeed, players belonging to the first cluster are normally characterized by statistical production on the low-end of the whole dataset, whereas moving towards higher-level clusters it is possible to note that statistical production also increases.

3.4 Further Analysis

In light of the results obtained in Section 3, it is possible to move onto a conclusive analysis on the newly-discovered groupings among NBA players.

Once the underlying clusters have been identified, it is possible to assess how players belong to a cluster differ from players of other clusters by analyzing the characteristics of each cluster. This implies focusing on the average traditional and advanced statistics of each cluster, as presented by Table 12:

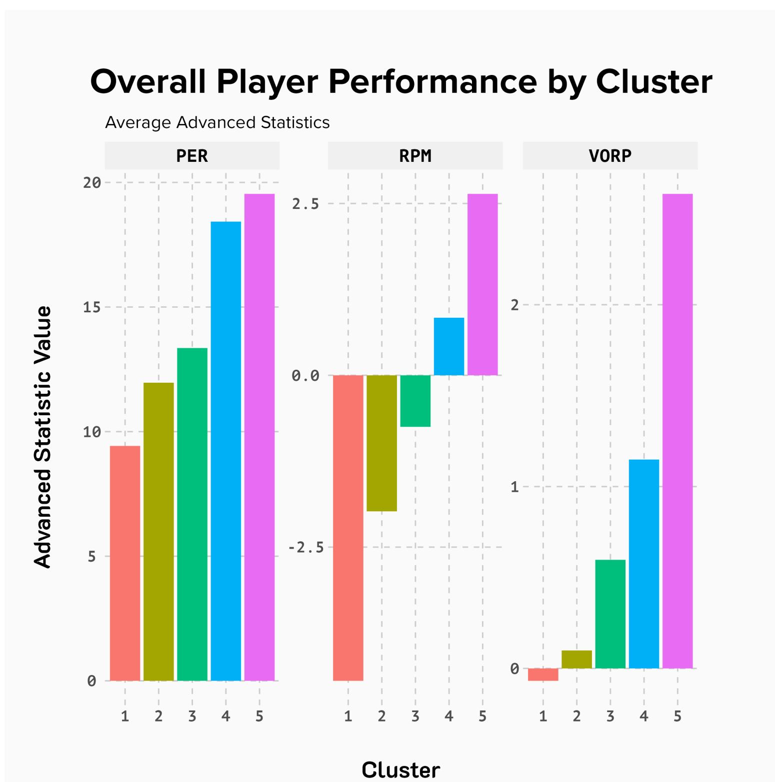

Figure 13: Plot displaying the average advanced statistics for each cluster

By assessing Fig. 13, it is possible to grasp the hierarchical partition which was evident before. Indeed, from the graph it appears that players belonging to cluster 1 perform the worst out of the five clusters according to each advanced statistic. This placement gradually increases while moving up the clusters, so that players within cluster 5 are characterized by the highest average advanced statistics, and therefore perform better, on average, than the players belonging to the other clusters.

The same kind of analysis can be focused on less-advanced statistics, as for instance the most classic three per-game statistics: points (`PTS_pg`), assists (`AST_pg`) and rebounds (`TRB_pg`). In this case, as Fig. 14 outlines, there is no longer a clear, increasing hierarchy within the observations. Indeed, it appears that players belonging to cluster 5 normally have a higher offensive production – as evidenced by the higher values of points per game (`PTS_pg`) and assists per game (`AST_pg`) – but fall behind when dealing with rebounding opportunities: players in cluster 4 normally record a higher rebounding average (`TRB_pg`). Conversely, another relevant insight is that players belonging to cluster 4 tend to underperform in terms of offensive production if compared to players in cluster 3.

Figure 14: Plot displaying the average traditional statistics for each cluster

An interesting point of analysis is that of assessing the composition, in terms of different positions, of each cluster, in order to identify meaningful explanation for the trends visualized previously. Specifically, our analysis will be focused on players who belong to clusters 3, 4 and 5. Starting by focusing on cluster number 3, it is possible to assess its composition by summarizing the total number of each position that belong to the cluster. By visualizing it through a pie chart presented in Fig. 15, it is possible to derive that a significant majority of this cluster is composed of Small Forwards – 34 out of 104 total components, 32.7%. Additionally, Shooting Guards and Point Guards each represent 24.0% of the total components. It follows that cluster 3 composed primarily by players that carry a significant portion of the offensive responsibilities. Specifically, players within this cluster rely primarily on the two-point shot, though they are also efficient from behind the three-point line. Additionally, these players often contribute to creating other offensive opportunities for their teammates, as highlighted by the high average of assists per game.

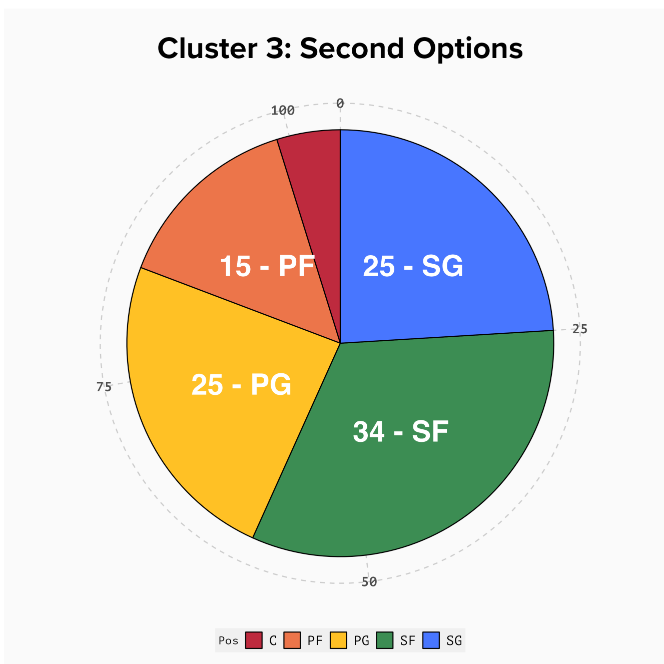

Figure 15: Plot displaying the composition of cluster 3, Second Options

Moving forward with the analysis of specific clusters, it is possible to focus on cluster 4 by assessing Fig. 15:

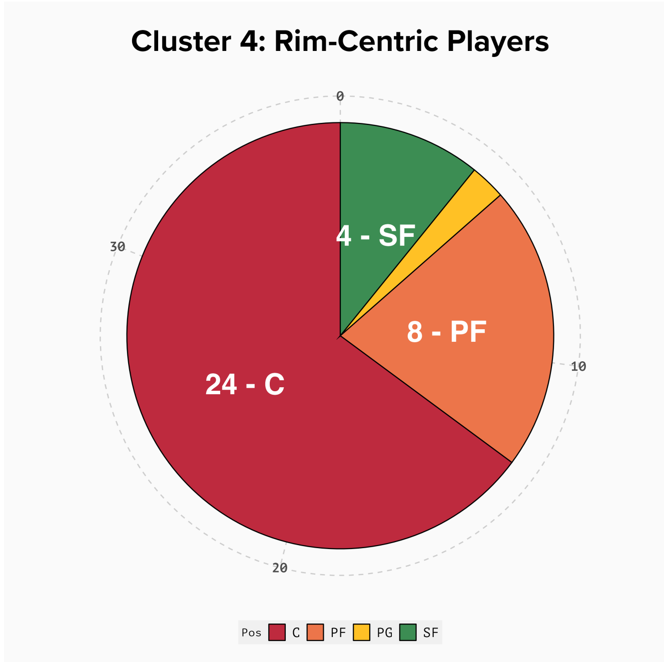

Figure 16: Plot displaying the composition of cluster 4, Rim-Centric Players

Fig. 16 highlights that a staggering majority of players belonging to this cluster are normally assigned a Center position – 24 times out of 37 (64.9%). The other prevalent position within the cluster is the Power Forward, which makes up 21.6% of the cluster. Conversely to cluster 3, in this case Point Guards appear only once within this cluster. Therefore, it appears that this cluster is primarily composed of players that occupy areas on the court nearest to the basket. Subsequently, players belonging to cluster 4 normally record a higher average of total rebounds as well as blocks. Additionally, the vast majority of their shooting attempts come from the two-point area rather than outside the three-point line. This translates to a higher average efficiency if compared to other positions, but also translates to lower average points scored or possessions created for other teammates.

Eventually, it is possible to analyze cluster 5 by assessing the primary positions that compose it, as expressed in Fig. 17:

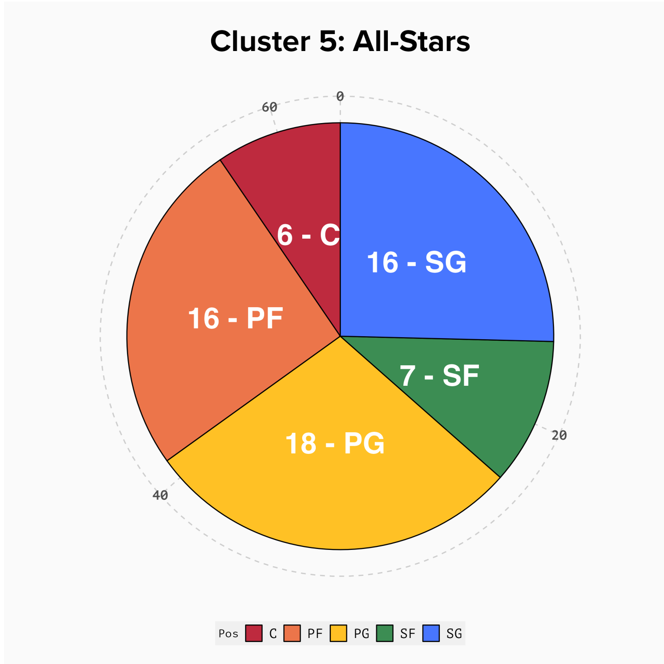

Figure 17: Plot displaying the composition of cluster 5, All-Star Players

As represented in Fig. 17, the position that majorly makes up this cluster is the Point Guard – 18 out of 63 times (28.6%). Moreover, one-half of the cluster comprehends Shooting Guards and Power Forwards, each with 16 observations out of 63 (25.4%). These players are characterized by the highest offensive production, if compared to all other players: their per-game statistics are almost always the highest of the groups (with an exception for assists and blocks, a prerogative of cluster 4); additionally, these players rely heavily on the use of the three-point field goal, which makes them a threat for contesting defenses. In turn, this increases the threat they represent, as their ability to shoot the ball allows for greater spacing and flow on offense. Their defensive contribution is limited in terms of blocks, primarily because these players are often the shortest on the floor; however, they make up for it thanks to their quickness and vision, that allows for a high average of steals per game. Ultimately, players belonging to cluster 5 are the ones whose offensive contributions are fundamental for their teams, and are thus the players that are more often on the court – 34 minutes per game on average.

4 Conclusion

Through Section 3.3 we developed a k-means clustering model which grouped NBA players in five major clusters, depending on their traditional and advanced statistics. Additionally, through Section 3.4 we assessed the major traits and characteristics of the three upper-tier clusters, those 3 through 5. It was thus possible to derive a partition of players in five major clusters whose composition looks like the following:

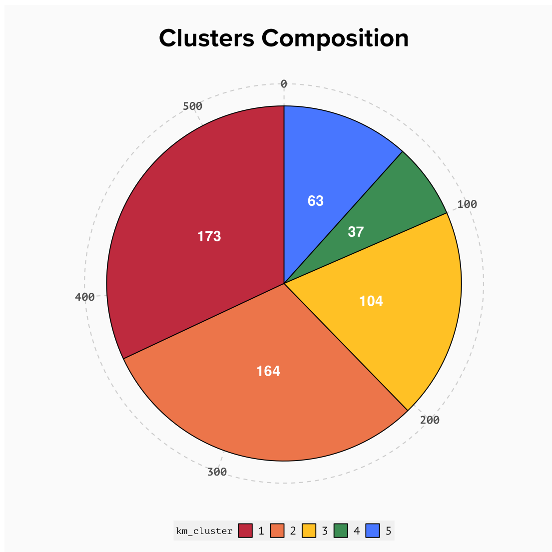

Figure 18: Plot displaying the number of players for each cluster

Additionally, thanks to the information and insights developed in Section 3 as a whole, it is possible to arrange such clusters in a clear hierarchy, thus leading to the introduction of new and fluid positions that allows to group players without regards for their effective position on the court, but rather by focusing on their offensive and defensive statistical production:

1. **Low-Tier Players**: a significant portion (32%) of players are concentrated in this cluster. These players are characterized by the lowest averages out of all clusters, and thus are the ones that least contribute to offensive and defensive production for a team. Their playing time is limited to few minutes per game, thus players belonging to this cluster may primarily be reserves who start playing towards the last minutes of regulation in already-closed games;

2. **Underperforming Players**: the number of players in this cluster is the second-highest, making up 30.3% of the total. These players are not statistically different from players within cluster 1, but in general under-performing players tend to record higher averages if compared to members of the previous tier. Their advanced statistics are still significantly low. Nonetheless, these players play substantially more than Low-Tier players (17.1 average minutes per game, compared to 7.6 of Low-Tier players);

3. **Second Options**: as already highlighted through Fig. 15, players belonging to this cluster are primarily Shooting Forwards and Guards. While the former category tends to rely heavily on the use of the two-point shot, the latter tends to shine particularly bright from three-point. This implies that players in this tier carry a significant portion of the offensive responsibilities, often having to create scoring opportunities not only for them but also for their teammates. Members of this category log the second-highest minutes per game, thus highlighting their relevance to their team's offense. Such players are therefore Second Options: they carry huge offensive responsibilities, despite not being elite-level players. This often translates to lower advanced metrics;

4. **Rim-Centric Players**: the number of players belonging to this cluster is the lowest, only 37 players (6.8% of the total). These players are characterised by a significant relevance and presence around the basket. Indeed, the majority of the scoring opportunities for these players come from two-point field goals. Additionally, they contribute significantly to creating second opportunities for their teams (they record a high average of offensive rebounds), while also protecting their basket from opponents' scoring opportunities, as highlighted by their high average of blocks per game. Rim-Centric players primarily benefit from their physical archetype, which allows them to be more effective round the rim both on offense and on defense;

5. **All-Stars**: players belonging to cluster 5 are the ones that perform better if compared to other players. They can be considered All-Stars. Indeed, such players always log higher average statistics if compared to their counterparts in other clusters, except for rebounds per game and blocks per game (a prerogative of cluster 4). All-Stars carry the majority of their teams' offensive responsibilities and are

particularly good at it. In turn, they log the highest minutes on average, and their teams perform significantly better with them on the court.

In light of what is presented above, it follows that the model was able to accurately group players according to their productivity, measured by traditional and advanced statistics. This highlights the similarities and differences between NBA players during a specific regular season, and also offers an insight into roster construction. Indeed, teams should aim at having a diverse roster in which players have different skill-sets and style of play so that they complement each other. This would ultimately contribute to team success.

4.1 Final Remarks

This paper allowed to cast light upon the use of analytics within sports by focusing on the National Basketball Association and its players.The assessment of historical trends within the NBA and the way basketball is played allowed to lay the introductory grounds on which the subsequent cluster analysis was developed.

In the last two decades there have been two major changes that can be analyzed to represent the evolution of the way basketball is played. The primary change is related to the speed at which teams play during games: *pace* has constantly increased in the last two decades, with an even steeper increase starting from the 2012-13 NBA Regular Season. This in turn translates to faster possessions for teams on offense, and therefore also to a higher number of possession during a game. This contributes to put a significant strain on the defensive team, while generating additional scoring opportunities for the offense. Moreover, another major trend that directly correlates to the evolution in pace is related to the three-point field goal. The use of this kind of shot has – as for pace – constantly increased in the last two decades, with a steep incline from the 2012-13 season. This highlights the clear correlation between the two factors: a faster offense allows to generate shots more quickly, thus straining the defense with additional scoring opportunities. At the same time, the reliance on the three-point shot contributes to spacing the floor and thus generating more fluid offense. In turn, a more fluid offense will flow at a faster pace, thus fostering the emergence scoring opportunities.

Therefore, it appears that the classical partition of players in five on-court position is no longer enough. Indeed, it is possible to partition players so that they are grouped with players whose performance – in terms of on-court statistics – is similar. Through such analysis, it appears that there exist five major groupings of players, depending on their performances in traditional stats as points or assists per game, as well as in more advanced stats, as Box Plus/Minus. Each of these five categories have their own characteristics, and can be placed on a vertical hierarchy. At the bottom of the ladder there is the first cluster – the one containing players whose performances are the lowest among the groups. Moving upwards on this hierarchy, it is possible to highlight a constant improvement in performances, up until the top of the ladder. This cluster corresponds to All-Stars, players that shine in every statistical category and that significantly contribute to their teams' success.

The analysis of historical trends helped discover important patterns and correlations among different aspects of the game of basketball. It allowed to drawn conclusions on the primary sources of change that shaped the way modern-day basketball is played. In addition, it was also possible to develop a model to assess players' similarities and differences according to a chosen criterion. The clusters that were identified were able to group players according to their skill-sets and productivity, measured with traditional and advanced statistics.

4.2 Further Applications

The following paper only represents one of the many possible applications of data analytics within sports. Specifically, clustering may be useful in assessing several more trends and looking for similarities between teams or players from different perspectives.

A further application of this clustering model may be taking into consideration additional players data, primarily in terms of contracts and salaries information. Specifically, it would be possible to compare some advanced statistics against player salaries, highlighting the clusters to which the observations belong. This would allow to identify players whose performances do not match their salaries, either because they are under-paid or over-paid.

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